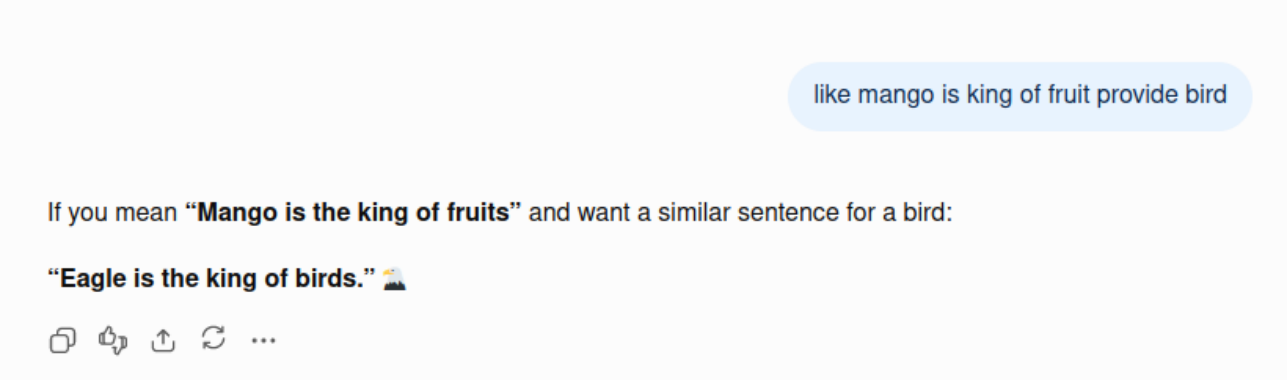
Lab 2: Types of Prompts and Prompt Engineering Basics

Objective: Learn types of prompts (instructional, interrogative, zero/few-shot).

Zero shot prompt:



One shote prompt:



Few shot prompt:

provide few more examples of it

Sure! Here are a few similar examples:

1. **Mango is the king of fruits.** 🥭
2. **Rose is the queen of flowers.** 🌹
3. **Eagle is the king of birds.** 🦅
4. **Lion is the king of animals.** 🦁
5. **Peacock is the king of beautiful birds.** 🦚
6. **Coconut is the tree of life.** 🥥
7. **Sun is the king of the sky.** ☀️
8. **Gold is the king of metals.** 🪙
9. **Tiger is the king of the jungle.** 🐅
10. **Apple is the queen of fruits.** 🍎

Asking for step-by-step reasoning:

Explain how to make pavbhaji step by step

🥘 How to Make Pav Bhaji — Step by Step

Ingredients:

- 2 potatoes
- 1 cup cauliflower
- ½ cup green peas
- 1 capsicum
- 2 tomatoes
- 1 onion
- 2–3 tbsp butter
- 1–2 tbsp pav bhaji masala
- ½ tsp turmeric powder
- ½ tsp red chilli powder
- Salt to taste
- 1 cup water
- Pav (bread rolls)
- Lemon and coriander for serving

Steps:

Lab 3: Precision Prompting for Information Extractio
Objective: Extract structured data from ChatGPT responses.

formating (like JSON, table, bullet points)

Here are examples of **formatting prompts** using JSON, tables, and bullet points:

1. JSON Format

Prompt:

> Give information about an eagle in JSON format.

Output:

JSON

```
{
  "name": "Eagle",
  "category": "Bird",
  "habitat": "Mountains and forests",
  "food": "Fish and small animals",
  "special_feature": "Excellent eyesight"
}
```

2. Table Format

Prompt:

> Give information about three birds in a table.

Bird	Habitat	Food	Special Feature
Eagle	Mountains	Fish, small animals	Sharp eyesight
Peacock	Forests	S…, insects	Beautiful feathers
Parrot	Forests	Fruits, seeds	Can imitate sounds

3. Bullet Point Format

Prompt:

> Give five facts about an eagle in bullet points.

- Eagle is a bird of prey.
- It has excellent eyesight.
- It has sharp claws.
- It can fly at high speeds.
- It builds nests in high places.

Lab 4: Summarization and Text Transformation.

Objective: Use ChatGPT for summarization, rewriting, tone transformation.

1. Summarize news articles, research abstracts:



Man Found Dead In Boat On Lake

INDIAN LAKE — A 53-year old Cincinnatian, vacationing with his wife, eight children and 18 grandchildren at the lake, suffered a fatal heart attack Tuesday as he rowed his boat after the motor had sheared a pin. He had been out on the lake only a little more than ten minutes.

Edmund F. Muenchen, a furniture dealer in the Queen City, was alone in the boat.

Dr. Frederick Kaylor, Logan County coroner, ruled 11:07 as time of death, resulting from a heart attack.

The victim was seen by Mrs. Paul Leary as he slumped over in the boat, drifting toward Paradise Island. He was only a short distance around a bend in the lake, north of his rented cottage at White Cottage Park. An early report, one of his grandchildren was with him, was quickly discounted. Lakeview fire department emergency squad, Lake Shore Patrol, sheriff's deputies and Dr. Kaylor arrived shortly after at the scene. Efforts to revive the man were futile.

Rev. Charles Conrad, Bellefontaine priest, was called to minister the last rites after was learned the Russells P priest was away from his par

There were no fishing pole the boat and relatives theori he was just going for a ride.

Dr. Kaylor said relatives him he had recently visited doctor, where he complained shoulder pains. He had been toring for a flu attack.

His body was returned to cinnati for final arrangements.

2 Rescue Crews Seek Lost Plane

GUAYAQUIL, Ecuador. (AP Two rescue parties struggled 15,423-foot-high Mt. Pichincha day, trying to reach the wreck of a U.S. military transport which it was feared 18 pers died.

Air searchers spotted the wre age 14,000 feet up the mount Tuesday and reported they saw sign of life.

Summarize news articles, research abstracts

Here is a concise summary of the **two news articles** shown in the image. A good summary should focus o the main event and key facts while leaving out minor details. The Classroom +1

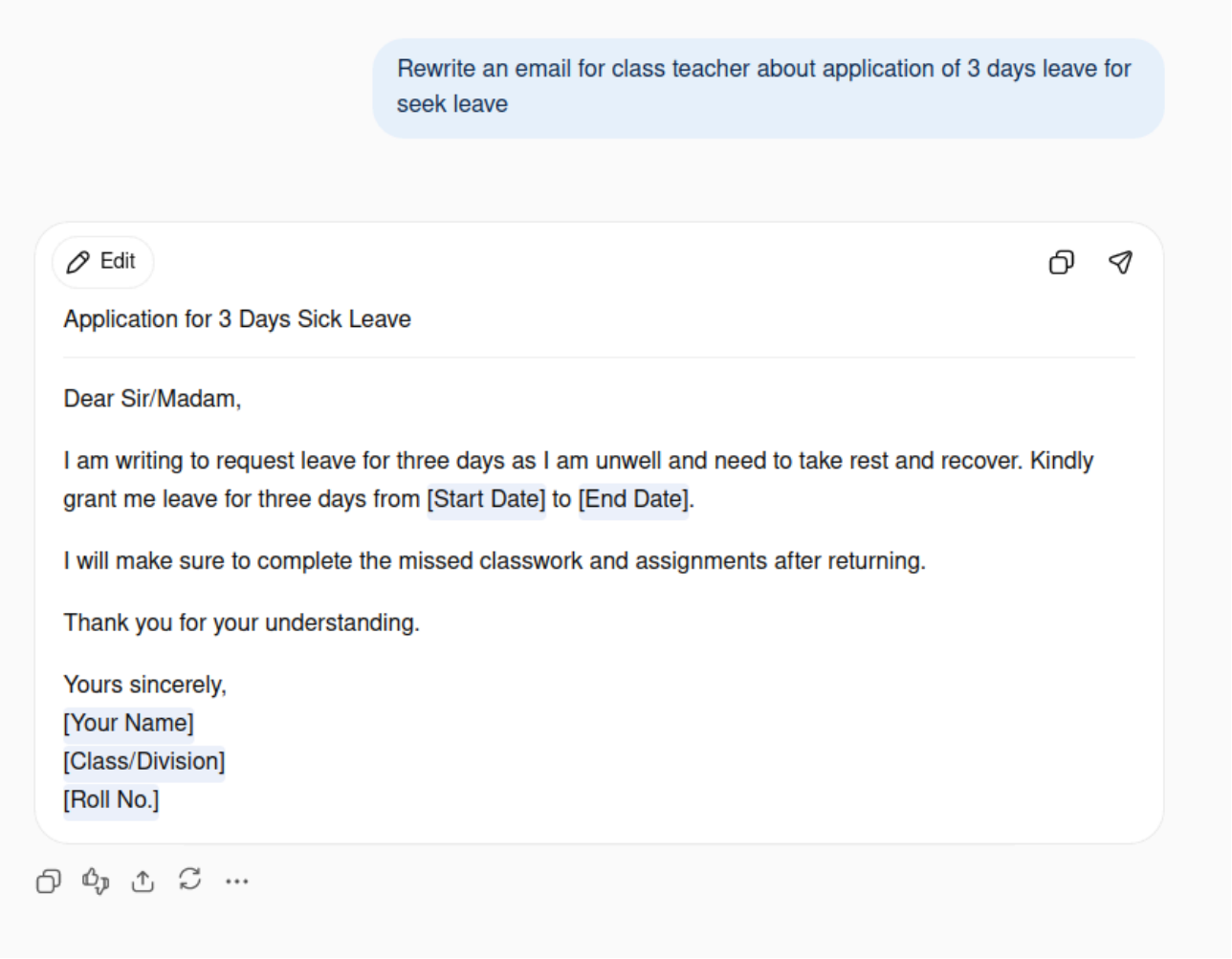
1. Man Found Dead in Boat on Lake

A **53-year-old Cincinnati physician, Edmund E. Mench**, was found dead in his boat on a lake. He had been rowing alone after the boat's motor stopped working because a pin had sheared. **Dr. Frederick Kaylo** the county coroner, determined that he died at about **11:07 a.m. from a heart attack**. His body was later recovered and returned to Cincinnati for funeral arrangements. Relatives believed he had gone out simply f a short boat ride.

2. Rescue Crews Seek Lost Plane

Two rescue teams in **Guayaquil, Ecuador**, climbed **Mount Pichincha** to reach the wreckage of a U.S. military transport plane. The aircraft was believed to have been carrying **18 people**, and there were fears that none survived. Air searchers had spotted the wreckage at about **14,000 feet** but reported seeing **no signs of life**. The rescue teams were attempting to reach the crash site despite difficult mountain condition

2. Rewrite emails in formal/informal tone :



Lab 5: ChatGPT in Code Generation and Debugging

Objective: Explore how ChatGPT can assist in programming.

1. Generate Python code from prompts:

Give me simGive me python code for finding factorial

Here is a simple Python program to find the **factorial of a number**:

Python Run

```python
num = int(input("Enter a number: "))

factorial = 1

for i in range(1, num + 1):
    factorial = factorial * i

print("Factorial of", num, "is", factorial)
```

Example output:

```
Enter a number: 5
Factorial of 5 is 120
```

Because **5! = 5 × 4 × 3 × 2 × 1 = 120**.

2.Debug a given code snippet using ChatGPT:

```python
# Student Marks Calculator

name = input("Enter your name: ")
marks1 = int(input("Enter marks for Subject 1: "))
marks2 = int(input("Enter marks for Subject 2: "))
marks3 = int(input("Enter marks for Subject 3: "))

total = marks1 + marks2 + marks3
percentage = total / 300 * 100

if percentage >= 75:
```

Show more

Here is the corrected code:

Python Run

```python
# Student Marks Calculator

name = input("Enter your name: ")
marks1 = int(input("Enter marks for Subject 1: "))
marks2 = int(input("Enter marks for Subject 2: "))
marks3 = int(input("Enter marks for Subject 3: "))

total = marks1 + marks2 + marks3
percentage = total / 300 * 100

if percentage >= 75:
    grade = "A"
elif percentage >= 60:
    grade = "B"
elif percentage >= 50:
    grade = "C"
elif percentage >= 40:
    grade = "D"
else:
    grade = "F"

print("Student:", name)
print("Total Marks:", total)
print("Percentage:", percentage)
print("Grade:", grade)
```

Error in your code

You wrote:

Python Run

```python
else
    grade = "F"
```

It should be:

Python Run

```python
else:
    grade = "F"
```

The `:` is required after `else` in Python.

Lab 6: Domain-Specific Applications

Objective: Use ChatGPT tasks.

Educational quiz generation:

